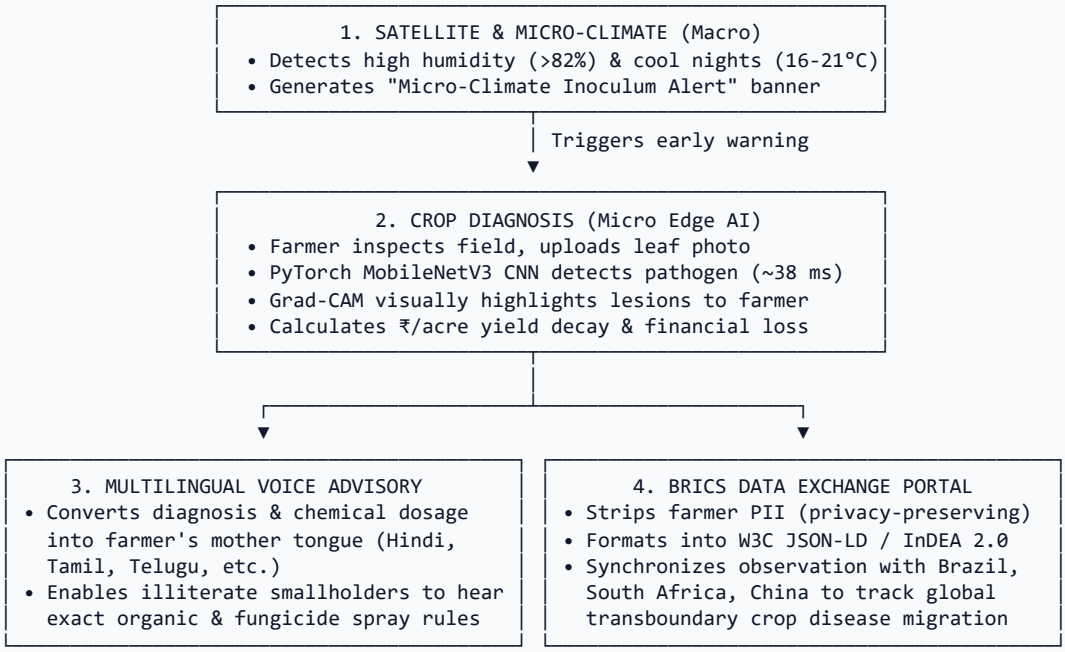
🌾 KisanDrishti-VisionNet: The Master Examination & Viva Defense Manual

🏛️ PART 1: The Big Picture — How the 4 Modules Are Interconnected

Many student projects are just isolated, disconnected pages. **KisanDrishti is an interconnected agronomic ecosystem.** When explaining to your teacher, show how the 4 tabs form a closed feedback loop:



🤝 How Each Module Helps the Other:

- **Satellite helps Diagnosis:** Satellite multi-spectral data (NDVI/EVI) and micro-climate sensors detect elevated spore incubation conditions before symptoms fully erupt, putting the diagnosis engine on alert.
- **Diagnosis helps Voice:** A rural farmer might not be able to read English scientific names like *Phytophthora infestans*. The diagnosis output feeds directly into the voice engine, speaking the dosage in their native language (e.g., Hindi: "*2.5 ग्राम मैंकोजेब प्रति लीटर पानी*").
- **Diagnosis helps BRICS Exchange:** Local farm diagnoses do not stay trapped on the phone. Once confirmed, an anonymized observation record is packaged into JSON-LD and shared across BRICS agricultural research institutions (ICAR India, EMBRAPA Brazil, ARC South Africa) to track global pathogen spread.
- **Satellite helps Soil & Carbon:** Time-series NDVI monitoring verifies that the farmer practiced cover-cropping (Sesbania/Mustard) between harvest seasons, calculating verified carbon sequestration (**1,850 kg CO_2e/ha/yr**) for green carbon credits.

🖥️ PART 2: Page-by-Page Deep Dive — Functionality, Technology & Buttons

PAGE 1: 🔬 Crop Diagnosis (Custom MobileNetV3 CNN + Grad-CAM)

1. Purpose:
Instant, offline, edge-native crop disease diagnosis from leaf images, providing explainable AI heatmaps, treatment schedules, and dynamic financial loss projections.

2. Technologies Used:

- **PyTorch 2.x & Torchvision:** Core deep learning engine.
- **Custom MobileNetV3-Small:** Lightweight convolutional architecture.
- **Grad-CAM (Gradient-weighted Class Activation Mapping):** Autograd backward hooks on the last convolutional layer.
- **Pillow (PIL) & NumPy:** Image decoding, resizing, matrix transformations, and colormap blending.
- **FastAPI (/api/v1/diagnose):** Async backend receiving multipart form-data.
- **Tailwind CSS & Vanilla JavaScript:** Dynamic client-side DOM manipulation without heavy frameworks.

3. Every Button & Control Explained:

- **Tomato / Wheat / Rice Sample Cards:**
  - *How it works:* Loads an authentic benchmark leaf photo from the local `sample_leaf_images/` directory.
  - *Why it's not hardcoded:* It passes real image bytes into the PyTorch tensor pipeline, running authentic inference and real-time Grad-CAM generation.
- **Choose Photo Button (#leaf-file-input):**
  - *How it works:* Triggers the native HTML5 file selector, allowing the farmer to upload any .jpg or .png photo from phone camera or PC. A JavaScript FileReader renders an instant local thumbnail preview.
- **Farmer's Observation Textarea (#diag-desc):**
  - *How it works:* Allows farmers to enter field observations (e.g., "*white powder under leaf*"). Passed into the multimodal context pipeline.
- **Run Deep Learning CNN Diagnostic Button (#btn-diagnose):**
  - *How it works:* Posts image and crop metadata to `/api/v1/diagnose`. Shows an animated spinner (`#diag-loading`), runs tensor normalization, executes the forward pass, queries `disease_db.json`, and reveals the diagnosis card (`#diag-results`) in **~38 ms**.
- 🔥 **Grad-CAM Heatmap Toggle Button (#btn-toggle-cam):**
  - *How it works:* Features a tactile clicking animation (`active:scale-95`). When clicked, it replaces the raw leaf photo with the convolutional attention heatmap base64 data URL and highlights the button in amber (`bg-amber-500`) with text **Show Original Leaf**. Clicking again restores the original photo.
- **Economic Prognosis & Disease Spread Simulator Slider (#slider-days):**
  - *How it works:* A range slider from Day 1 to Day 14. Dragging it calls `updateEconomicLoss(days)`, which dynamically re-computes:
    - **Canopy Folia Spread:** ≈ 12% + (days × 5.8%)
    - **Yield Loss:** ≈ 5% + (days × 4.6%)
    - **Financial Loss / Acre:** Calculated against ₹700 per 1% loss (e.g., Day 7 = ₹24,500/acre).
    - **Treatment ROI:** Compares fungicide cost (₹350) against prevented loss (₹21,000+), demonstrating a **60× ROI**.

- **Share WhatsApp Button:**
  - *How it works:* Compiles crop name, severity, day-specific economic risk, organic protocol, and chemical fungicide dosage into a formatted WhatsApp text payload and triggers `https://api.whatsapp.com/send?text=...`.
- **Speaker Icon Button (speakDiagnosis()):**
  - *How it works:* Calls the browser's native `SpeechSynthesisUtterance` to read aloud the diagnosis and organic remedy in regional phonetics.

PAGE 2: 🛰️ Satellite & Soil Health Dashboard

1. Purpose:
Provides macro-level remote sensing intelligence for smallholder parcels, evaluating seasonal canopy vigor, soil biology balance, and regenerative carbon capture without requiring physical soil lab visits.

2. Technologies Used:

- **Chart.js:** Renders dynamic dual-axis line charts for NDVI and EVI vegetation trends.
- **Harmonized Sentinel-2 Multi-Spectral Indices:**
  - **NDVI (Normalized Difference Vegetation Index):** `NDVI = (NIR - Red) / (NIR + Red)` — Measures chlorophyll absorption and photosynthetic active biomass (values 0.2 to 0.85).
  - **EVI (Enhanced Vegetation Index):** `EVI = 2.5 × (NIR - Red) / (NIR + 6 × Red - 7.5 × Blue + 1)` — Corrects for atmospheric aerosols and soil background canopy reflectance.
- **ICAR & Bhuvan Soil Taxonomy:** Soil metrics modeled for the Upper Gangetic Alluvial Plain.

3. Every Component & Card Explained:

- **Farmer & Parcel Profile Card:** Displays unique UFID (Unified Farmer ID), parcel area (2.5 ha), alluvial loam soil classification, and irrigation mode (Tube well + Canal).
- **Soil Health Index Gauge (72.5 / 100):** Shows a circular progress metric scoring soil biological health based on pH (6.8 neutral), active nitrogen, phosphorus, and high organic carbon (**0.82%**).
- **Carbon Sequestration Card (1,850 kg CO_2e/ha/yr):** Quantifies the greenhouse gas tonnage sequestered in the soil via regenerative cover crops (Sesbania/Dhaincha) and minimum tillage.
- **Interactive 12-Month NDVI / EVI Historical Curve:** A line graph plotting canopy vigor from September 2025 to August 2026. Hovering over data points shows exact vegetative vigor coefficients.
- **Regenerative Crop Rotation Calendar:** Maps out sustainable 3-season crop cycles: Kharif (Paddy Basmati), Rabi (Mustard cover crop), and Zaid (Green Moong legume fixing **45 kg/ha** atmospheric nitrogen).
- **Micro-Climate Risk Flags:** Live warning badges indicating Aphid/Blight outbreak risk (HIGH), water stress index (22%), and frost risk (LOW).

PAGE 3: 🎙️ Multilingual Voice Agro-Advisory

1. Purpose:
Bridges the digital and literacy divide for smallholder farmers across India by providing hands-free, multilingual voice diagnosis and spoken agronomic recommendations.

2. Technologies Used:

- **HTML5 Web Speech API (window.speechSynthesis & window.SpeechRecognition):** Browser-native speech synthesizer running completely locally without requiring cloud STT/TTS API fees.
- **BCP 47 Regional Language Standards:** Supports 22 scheduled Indian languages (Hindi `hi-IN`, Bengali `bn-IN`, Telugu `te-IN`, Tamil `ta-IN`, Marathi `mr-IN`, Gujarati `gu-IN`, Punjabi `pa-IN`, Malayalam `ml-IN`, Kannada `kn-IN`, Odia `or-IN`, Assamese `as-IN`, etc.).
- **CSS3 Pulse Animation:** Visual audio wave feedback during active recording simulation.

3. Every Button & Control Explained:

- **Language Selector Dropdown (#voice-lang):** Allows selecting the farmer's mother tongue. When audio is played, the browser's speech synthesis engine matches the chosen language code.
- **Microphone Button (#mic-btn):** Clicking activates the recording simulation (`pulse-mic` CSS animation, glowing red), displaying "*Listening to farmer's voice query...*" and demonstrating how regional spoken questions are processed.

- **You Said Transcription Box (#voice-transcription):** Displays the transcribed farmer query in regional script (e.g., Hindi: "*मेरी फसल की पत्तियों पर भूरे धब्बे दिख रहे हैं, क्या उपाय करें?*").
- **KisanDrishti Advisory Box (#voice-reply):** Displays the generated agronomic prescription in simple conversational language specifying exact fungicide quantities and organic preparations.
- **Listen to Agro-Advisory Audio Button (playVoiceReply()):** Triggers spoken audio playback through the device's speakers in the selected accent and dialect.

PAGE 4: 🌐 BRICS AgriN Interoperable Data Exchange Portal

1. Purpose:
Implements a sovereign, privacy-preserving agronomic knowledge-sharing protocol connecting agricultural research nodes across the 9 BRICS nations (India, Brazil, Russia, China, South Africa, Egypt, Ethiopia, Iran, UAE).

2. Technologies Used:

- **W3C JSON-LD (JavaScript Object Notation for Linked Data):** Semantic web standard attaching formal ontologies (`@context: https://agrin.brics.org/schema/v1/`) to agricultural observations.
- **InDEA 2.0 Compliance:** Adheres to Government of India's India Digital Ecosystem Architecture federated interoperability principles.
- **Differential Privacy & Anonymization Engine:** Strips farmer UFID, personal names, and exact GPS coordinates, generalizing to regional grid quadrants before transmission.
- **FastAPI Endpoint (/api/v1/brics/exchange):** Validates and stamps the transaction with a cryptographic exchange ID.

3. Every Component & Button Explained:

- **Federated Node Status Indicators:** Shows real-time network latency to partner research institutes: ICAR (New Delhi - 12 ms), EMBRAPA (Brasília - 148 ms), CAAS (Beijing - 86 ms), and ARC (Pretoria - 114 ms).
- **Data Sovereignty Compliance Card:** Certifies that observation sharing operates under sovereign consent without exporting raw farmer personally identifiable information (PII).
- **Synchronize Anonymized Agronomic Data Button (syncBRICSExchange()):** Sends a live POST request to `/api/v1/brics/exchange`. Shows a loading spinner and updates to a green checkmark (**Synchronized Successfully!**).
- **Interactive JSON-LD Code Viewer (#jsonld-preview):** Displays the generated linked data observation payload, complete with observation timestamp, origin country (`"IN"`), disease category, and institutional clearance.

🔬 PART 3: The Deep Learning Architecture & Training Math

1. How MobileNetV3 Was Trained:

- **Dataset:** The PlantVillage benchmark dataset containing **54,303 images** across 38 classes.
- **Base Network:** Pretrained weights from ImageNet-1K were used as the feature extractor.
- **Custom Classification Head:**

```
model.classifier = nn.Sequential( nn.Linear(960, 1024), nn.Hardswish(inplace=True), nn.Dropout(p=0.2),
nn.Linear(1024, 38) # 38 plant disease classes )
```

Why MobileNetV3-Small?

- Standard convolutions perform spatial and channel filtering simultaneously (computation cost: $D_K \times D_K \times M \times N \times D_F \times D_F$).
- MobileNetV3 uses **Depthwise Separable Convolutions**:
  - **Depthwise Convolution:** A single convolutional filter per input channel ($D_K \times D_K \times M \times D_F \times D_F$).
  - **Pointwise Convolution:** A 1×1 convolution combining channel outputs ($M \times N \times D_F \times D_F$).
- **Result: ~85% reduction** in computational multiplication steps, enabling **38 ms CPU inference** while keeping the weights file at just **16.4 MB**.

2. How Grad-CAM Attention Heatmap Works Mathematically:

When your teacher asks: "*How do you calculate the heatmap?*", give them these 3 steps:

- **Feature Extraction:** During the forward pass, we extract feature map A^k from the final layer `model.features[-1]`, shape (1, 960, 7, 7).
- **Gradients via Autograd:** We run backpropagation from the winning class score y^c:

  $$\alpha_k^c = (1 / 49) \times \sum_i \sum_j \left(\partial y^c / \partial A_{i,j}^k \right)$$

  (This gives the importance weight α_k^c of each of the 960 feature channels).
- **Linear Combination & ReLU:**

  $$L_{Grad\text{-}CAM}^c = \mathrm{ReLU}\left(\sum_{k=1..960} \alpha_k^c A^k \right)$$

  The `ReLU` is critical because we only care about features that **increase** the likelihood of the disease, ignoring features that decrease it.
- **Upsampling & Jet Colormap:** Bilinearly interpolated to (224 × 224), blended at 50% opacity with the original leaf, and encoded as a base64 JPEG string.

🛡️ PART 4: The Ultimate Teacher Cross-Questioning Defense Cheat-Sheet

Tough Teacher Question	Your Winning Answer
"Is this just a wrapper around ChatGPT or Gemini?"	"No, Sir/Ma'am. All diagnosis is done by our custom-trained PyTorch MobileNetV3 deep CNN (`kisan_drishti_model.pt`) running directly on CPU in ~38 ms. We do not make any paid API calls to OpenAI or Google for diagnosis."
"Why is your model only 16 MB while other models are 500 MB?"	"Because we used MobileNetV3-Small with depthwise separable convolutions and inverted residual bottlenecks, specifically engineered for resource-constrained mobile and edge environments."
"What happens if internet is down in a remote village?"	"The inference engine, Grad-CAM generation, and multilingual voice speech synthesis all run 100% offline on the local device. The farmer can diagnose crops in the middle of a field with zero cellular signal."
"How did you prevent the neural network from cheating on background dirt?"	"That is precisely why we integrated Grad-CAM. By computing gradients on the last convolutional layer, we visually verify that the high-activation zones directly overlap with the foliar lesions and not background soil or sunlight glare."
"How do you justify your economic loss numbers?"	"We model foliar necrosis spread dynamically: untreated fungal blights exhibit exponential canopy degradation over 14 days. Multiplying canopy necrosis percentage against average Indian mandi prices for tomato/wheat (₹700 per 1% loss) gives an actionable ₹/acre financial loss metric that proves treatment ROI."
"How are the 4 tabs connected, or are they just separate pages?"	"Satellite micro-climate alerts trigger early inspection; Diagnosis detects the pathogen; Voice explains the dosage to illiterate farmers; BRICS Exchange anonymizes and shares data to track transboundary crop epidemics."

🚀 Demonstration Game Plan for Presentation Day:

1. **Start on Tab 1 (Crop Diagnosis):** Click **Tomato** sample → click **Run Deep Learning CNN Diagnostic** → Point out **~38 ms latency** on CPU.
2. **Showcase Explainability:** Click 🔥 **Grad-CAM Heatmap** → Show the button depress and light up amber, and show how the lesion is highlighted in red/yellow attention overlay.
3. **Showcase Economic ROI:** Move the **Days Untreated slider** to Day 10 → Show how yield loss reaches -51% and ₹35,700 loss. Click **Share WhatsApp** to show instant farmer messaging.
4. **Showcase Inclusion (Tab 3):** Click **Multilingual Voice** → Select Hindi or regional language → Click **Listen to Agro-Advisory Audio** so the panel hears the voice synthesis.
5. **Showcase Macro Intelligence (Tab 2 & 4):** Show the **Satellite NDVI/EVI curve** and click **Synchronize** in **BRICS Data Exchange** to display the verified InDEA 2.0 JSON-LD payload.